204 HTTP Status Code in ASP.NET Core Web API

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In this article, I will discuss **How to Return 204 No Content HTTP Status Code from the ASP.NET Core Web API Controller Action Methods**. Please read our previous article discussing **How to Return 200 Succeeded HTTP Status Code in ASP.NET** Core Web API with Examples.

204 HTTP Status Code:

The 204 HTTP Status Code stands for "No Content". It indicates that the server has successfully processed the request but has no content to return in the response body. This status code is commonly utilized in scenarios where a client sends data to the server (like a PUT or PATCH request), and the server processes the request successfully but does not need to send any data in the response back to the client.

For example, if a user updates their profile on a RESTful API, the server might respond with a 204 to the client to indicate that the update was successfully made, but there is no additional information to return. The following are some of the common scenarios where using the 204 status code is appropriate:

- **Successful PUT Requests:** When updating a resource on the server and no response body is needed.
- **Successful DELETE Requests:** When a resource has been successfully deleted from the server, no content is needed.
- **PATCH Requests:** When a PATCH request has been processed, but no additional content is required.
- **HEAD Requests:** The 204 status can be used for HEAD requests where the headers are returned without a response body.

How to Return 204 HTTP Status Code in ASP.NET Core Web API?

In ASP.NET Core Web API, returning HTTP Status Codes is useful in scenarios where an operation is performed on the server (like a DELETE, PUT, or a PATCH operation) that doesn't produce a response body and you want to indicate to the client that the request was successfully processed.

To return a 204 Status Code from an ASP.NET Core Web API controller, we need to use the **NoContent()** method. There only is one way to return a 204 status code from an ASP.NET Core Web API controller. The syntax to return a 204 status code from an ASP.NET Core Web API controller is as follows:

Using NoContent Method to Return 204 Status Code

Let us understand the 204 HTTP Status Code in an ASP.NET Core Web API with an example. First, create the following controller and action methods in a file named Employee.cs and add them to an existing or a new controller file.

```
[HttpPut]
public IActionResult Put(int id, Employee emp)
{
    if (id != emp.Id)
        return BadRequest("Mismatch between the provided id and employee data.");
    var employeeToUpdate = Employees.FirstOrDefault(e => e.Id == id);
    if (employeeToUpdate == null)
        return NotFound();
    // Update the properties of the existing employee with the new data
    employeeToUpdate.Name = emp.Name;
    employeeToUpdate.Position = emp.Position;
    employeeToUpdate.Salary = emp.Salary;
    // Return 204 No Content status code to indicate successful update
    return NoContent();
}
[HttpDelete("{id}")]
public IActionResult Delete(int id)
{
    var employeeToDelete = Employees.FirstOrDefault(e => e.Id == id);
    if (employeeToDelete == null)
        return NotFound();
    // Remove the employee from the list
    Employees.Remove(employeeToDelete);
    // Return 204 No Content status code to indicate successful deletion
    return NoContent();
}
```

In the above example,

- The EmployeeController method attempts to update an employee if the operation is successful, it returns a 204 No Content status code using the NoContent() method. The method also checks if the provided id matches the employee's Id. If a mismatch occurs, it returns a 400 Bad Request with a message indicating the mismatch.
- The EmployeeController method attempts to delete an employee. If the deletion is successful, it returns a 204 No Content status code using the NoContent() method. If the employee cannot be deleted (for example, it's not found), a 404 Not Found response is returned.

Testing the APIs:

Now, run the application and test the APIs using Postman, Fiddler, Swagger, or Slap. Here, You can test the APIs using Postman and Swagger. Testing using Postman provides more options and capabilities for sending

HTTP Returning 204 Status Code After Successfully Update Operation:

URL: https://localhost:7195/api/Employee/1
HTTP Method: PUT
Request Body:

Response Body: The Response body will be empty with 204 HTTP Status Code

HTTP Returning 204 Status Code After Successfully Delete Operation:

URL: https://localhost:7195/api/Employee/1
Method: DELETE

Response Body: The Response body will be empty with a 204 HTTP Status Code

Manual Status Code Setting:

Alternatively, you can use the StatusCode method to return a 204 Status Code instead of the NoContent method. You usually use StatusCode controller, as it allows you to set the StatusCode method to return a 204 Status Code.

```
[HttpPut]
public IActionResult Put(int id, Employee emp)
{
    if (id != emp.Id)
        return BadRequest("Mismatch between the provided id and employee data.");
    var employeeToUpdate = Employees.FirstOrDefault(e => e.Id == id);
    if (employeeToUpdate == null)
        return NotFound();
    // Update the properties of the existing employee with the new data
    employeeToUpdate.Name = emp.Name;
    employeeToUpdate.Position = emp.Position;
    employeeToUpdate.Salary = emp.Salary;
    // Return 204 No Content status code to indicate successful update
    return StatusCode(204);
}
[HttpDelete("{id}")]
public IActionResult Delete(int id)
{
    var employeeToDelete = Employees.FirstOrDefault(e => e.Id == id);
    if (employeeToDelete == null)
        return NotFound();
    // Remove the employee from the list
    Employees.Remove(employeeToDelete);
    // Return 204 No Content status code to indicate successful deletion
    return StatusCode(204);
}
```

With the above changes in place, save the application, and it should work as expected.

When to Use 204 Status Code in ASP.NET Core Web API?

In ASP.NET Core Web API, you should use the 204 Status Code in the following scenarios:

- **Successful Deletion of a Resource:** When a client sends a DELETE request to remove a resource, and there is no need to return any data.
- **Successful Update with No Content:** When a client sends a PUT or PATCH request to update a resource, and the data needs to be received after the update.
- **Successful Operation with No Response Data:** When a client sends a POST request (e.g., via SubmitForm) and the operation is successful but does not need to return any data.
- **Successful Search Operations:** When performing batch operations where multiple resources are processed, and no specific data needs to be returned to the client.
- **Action Methods with No Content:** In scenarios where a response is expected and there is no specific data to send, a 204 status code can be used.

In the next article, I will discuss **How to Return 201 HTTP Status Code in ASP.NET Core Web API with Examples**. In this article, I try to explain **How to Return 204 No Content HTTP Status Code in ASP.NET Core Web API** with Examples and I hope you enjoy the article. If I write the Content-HTTP Status Code in the ASP.NET Core Web API

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